

EEE 419/591

Summer 2026

Homework 3

[Same instructions as HW 1. Significant changes are highlighted in bold font]

Solve the following problem. **Read the instructions of the problem on the next page to know what you need to submit.** Submit your work to the designated link on Canvas. If you upload a second attempt to Canvas and it renames your file, ignore the renaming.

In this HW, only limited AI is allowed. Examples of allowed prompts on this HW:

- What is a greatest common divisor of two numbers?
- How do I write an if statement in python? Give me an example
- How do I check if a number is integer in python?
- I got this error message. How do I interpret it?

Examples of prohibited prompts on this HW:

- Write me a code that calculates the greatest common divisor of a list of numbers in python
- I ran this code. [PASTE CODE HERE]. I got this error message. [PASTE THE ERROR MESSAGE HERE]. How do I fix my code?

Do not forget to cite your AI chat if you used any. Check the Syllabus document for details.

In this homework problem you will learn to stamp components from a circuit into a matrix describing that circuit using the methods from nodal analysis in Circuits 1. Make sure to test your code with a simple circuit problem. Canvas has three attachment python files. Use them to finish this homework assignment.

The code you will submit should print two things: a single vector and a number.

- If the circuit has only current sources, then
 - the vector printed will contain the node voltages.
 - the number printed will contain the average of these voltages
 - Example:

```
Vector is [3.  1.5  1.5  3.]
```

```
Node voltages' average is 2.25 Volts. There are no voltage  
sources in the circuit
```

- If the circuit has one, or more, voltage source(s), then
 - the vector printed will contain not only node voltages but also the actual current passing through each of the present voltage source(s) in the circuit. Remember that the current value(s) will be the last element(s) in the vector, if you follow the method taught in the lecture videos, which is what is highly recommended.
 - the number printed will be the average of the voltages in that vector (i.e. do not include the currents in your average calculation).
 - Example:

```
Vector is [3.  1.5  1.5  3.  -4.5  -4.5]
```

```
Voltages average is 2.25 Volts
```

Notes and Hints:

- You should submit hw3.py. Do not submit any of the below files (other than your, significantly modified, hw3.py)
- You may use the schematics in the lecture as tests to see if your solution is correct. However, the grader will also run your code on other circuits which may have multiple current sources, voltage sources, or both, and far more resistors. So, make up some circuits of your own for which you can easily compute the answers to check your work.
- You may assume that the nodes will be numbered 0-N consecutively so you need not worry about missing node numbers.
- Important Hint: Do not write a code that resizes the arrays midway! You should be able to write a code that sizes the arrays correctly at the beginning of the code before starting to populate these arrays!

MANDATORY – Reminder to cite AI chat – Check syllabus document for details